# COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 8 : SLO – 2 | Binary arithmetic and BCD operations
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## I. Match the Following

• 1. Sign bit in binary number — b) 1 for negative and 0 for positive
• 2. 1's complement of 0101 — a) 1010
• 3. 2's complement of 1001 — e) 0111 (0110 + 1)
• 4. Binary division rule — c) Shift and subtract process
• 5. BCD for decimal 42 — d) 0100 0010

## II. Fill in the Blanks

• 1. 0 2. 1101 3. 1 4. decimal long 5. 4

## III. Activity / Describe and Perform

• 1. Sign Magnitude — (+4)+(–2): different signs → 100–010 = 010, sign of larger (+) → 0010 = +2. (–4)–(+2) = –(4+2) = –6 = 1110.
• 2. 1's complement subtraction, 11–5: 1011 + 1's of 0101 (1010) = 1 0101 → end-around carry → 0101+1 = 0110 = +6.
• 3. 2's complement subtraction, 6–3: 0110 + 2's of 0011 (1101) = 1 0011 → discard carry → 0011 = +3.
• 4. Multiplication 110 × 11 (6×3): 110 + 1100 (shifted) = 10010 = 18. Division 1100 ÷ 11 (12÷3): long division → quotient 0100 = 4, remainder 0.
• 5. BCD: 42 = 0100 0010. BCD addition 27+35: 0010 0111 + 0011 0101 = 0101 1100; units 1100 invalid → +0110 → 0110 0010 with carry → = 62 (0110 0010).